

Contoso Limited

Infrastructure Documentation

Created by Lead Systems Engineer



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INTRODUCTION

Critical infrastructure data has been collected to get a comprehensive picture of the Contoso Ltd network. The capture consisted of device patching, wiring and general connectivity. It also included physical rack layout and a production hardware summary. Basic switching and routing were also included as so was application services like DHCP, DNS and more.

The collected data have been analyzed, streamlined and put into this document to present a detailed overview of the network intending to provide a better understanding of its strength, its weaknesses and in general to help make better decisions around improving the overall health of the network.

Good to Know

The (Work in progress / example only) reference represents content that are still considered work in progress typically showing as incomplete tables, diagrams or headings.

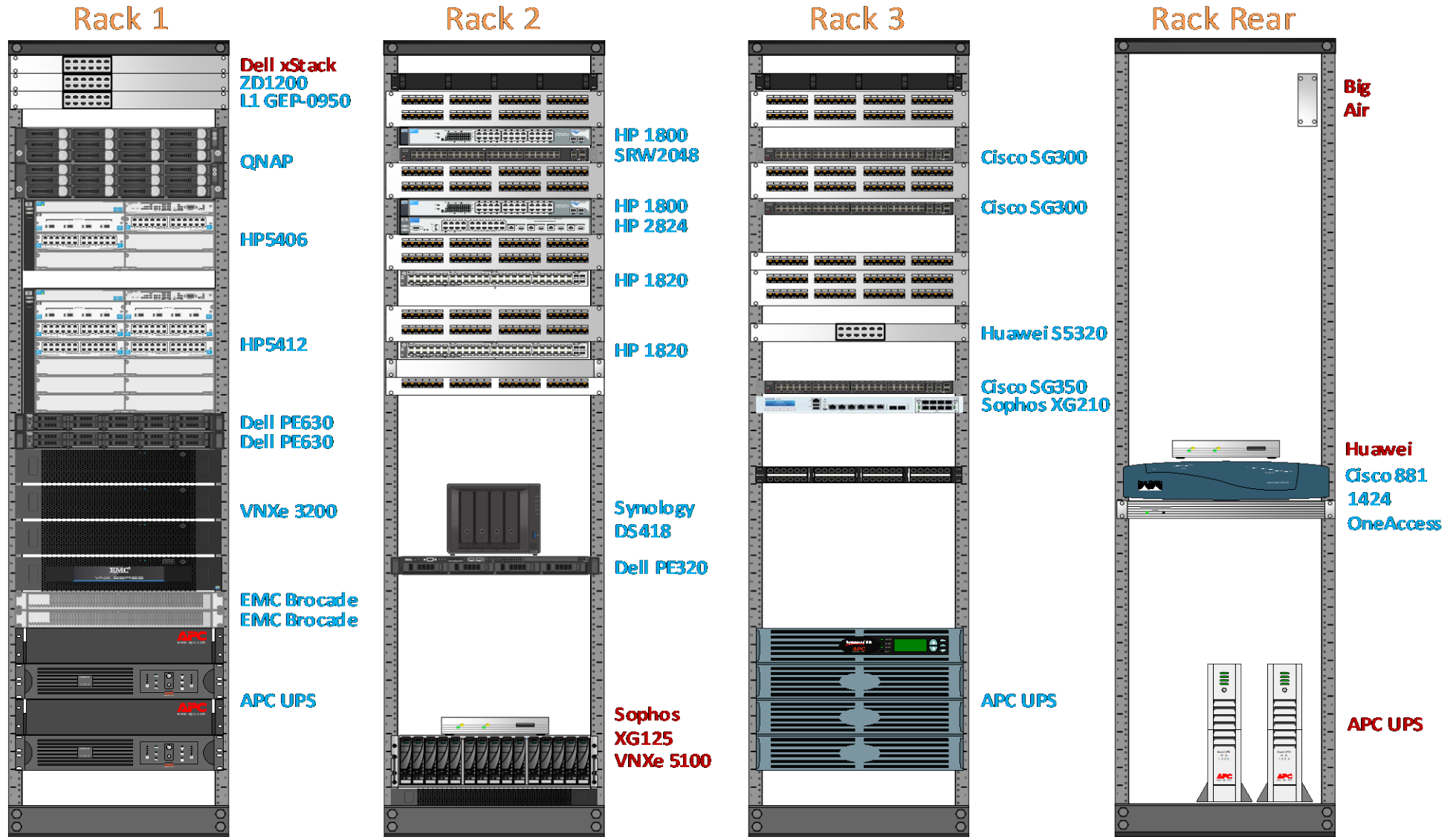
There are multiple references to data in tables that are either colored red or are colored ~~red with strikethrough~~. Most of these highlights reference items that should be considered obsolete, outdated, decommissioned or out of service.

There are also multiple references to "clients" which in this document typically means client Workstations.

RACKS AND HARDWARE

Below diagram illustrates the layout of the Contoso Ltd server room and describes the hardware currently deployed. There are 3 production racks in use and most of them contain live production equipment.

Devices marked in red are either powered off, not in use or are in the process of being decommissioned.



DEVICE CAPTURE

The purpose of tables below is to present a summary of each active devices in the network and to be able to quickly determine the type, role and location.

Certain devices have multiple IPs assigned which is not necessarily represented in the list, for further detail please refer to the [VLAN Mapping section](#) for devices with multiple IPs.

Layer 2 and 3 Devices

Device	Device Type	Location	IP Address
1424 OneAccess SHDSL	Router	Serverroom Rack 3 Back	-
Cisco 881	Router	Serverroom Rack 3 Back	10.5.1.211
Cisco SG300-52P	Switch PoE Layer2 & 3	Serverroom Rack 3 Middle	10.5.1.219
Cisco SG300-52P	Switch PoE Layer2 & 3	Serverroom Rack 3 Top	10.5.1.237
Cisco SG350-52P	Switch PoE Layer2 & 3	Serverroom Rack 3 Bottom	10.5.1.221
Dell xStack DES-3200-10	Switch Layer 2	Serverroom Rack 1	?
EMC Brocade DS-300B	Switch Fibre Channel	Serverroom Rack 1	10.5.1.238
EMC Brocade DS-300B	Switch Fibre Channel	Serverroom Rack 1	10.5.1.239
HP Aruba 2530 24G	Switch Layer 2	Large Conference Room	10.5.1.32
HP ProCurve 1800-24G DS1	Switch Layer 2	Serverroom Rack 2 Bottom	10.5.1.233
HP ProCurve 1800-24G DS2	Switch Layer 2	Serverroom Rack 2 Top	10.5.1.234
HP ProCurve 2824 DS3	Switch Layer 2	Serverroom Rack 2	10.5.1.232
HP ProCurve 5406zl	Switch Layer 3	Serverroom Rack 1 Top	10.5.1.240
HP ProCurve 5412Rzl2	Switch Layer 3	Serverroom Rack 1 Bottom	10.5.1.210
HPE OfficeConnect 1820	Switch Layer 2	Serverroom Rack 2 Bottom	10.5.1.236
HPE OfficeConnect 1820	Switch Layer 2	Serverroom Rack 2 Top	10.5.1.241
Huawei S5320	Network Termination Unit	Serverroom Rack 3	-
Huawei noname	Network Termination Unit	Serverroom Rack 3 Back	-
Level One Switch GEP-0950	Switch PoE Layer2 & 3	Serverroom Rack 1	10.5.1.1
Linksys Switch SRW2048	Switch Layer 2	Serverroom Rack 2	10.5.1.235
Wireless AP122	Wireless Access Point	Open floor ceiling	10.5.1.122
Wireless AP162	Wireless Access Point	Open floor ceiling	10.5.1.162
Wireless AP176	Wireless Access Point	Open floor ceiling	10.5.1.176
Wireless AP178	Wireless Access Point	Open floor ceiling	10.5.1.178
Wireless AP181	Wireless Access Point	Open floor ceiling	10.5.1.181
ZoneDicator 1200	Wireless Controller	Serverroom Rack 1	10.5.1.209
Sophos XG 210	Router / Firewall	Serverroom Rack 3	10.5.1.252

Servers

Device	Device Type	Location	IP Address
Dell PowerEdge R320	Enterprise Physical Server	Serverroom Rack 2	10.5.1.6
Dell PowerEdge R630 Mgmt	Enterprise Physical Server	Serverroom Rack 1	10.5.2.100
Dell PowerEdge R630 iDrac	Enterprise Physical Server	Serverroom Rack 1	10.5.1.230
Dell PowerEdge R630 vMotion	Enterprise Physical Server	Serverroom Rack 1	10.0.70.10
Dell PowerEdge R630 Mgmt	Enterprise Physical Server	Serverroom Rack 1	10.5.2.101
Dell PowerEdge R630 iDrac	Enterprise Physical Server	Serverroom Rack 1	10.5.1.231
Dell PowerEdge R630 vMotion	Enterprise Physical Server	Serverroom Rack 1	10.0.70.11

NAS Storage

Device	Device Type	Location	IP Address
QNAP NAS NAS01	Network Attached Storage	Serverroom Rack 1	10.5.1.207
Synology NAS-DR	Network Attached Storage	Serverroom Rack 2	10.5.1.128
Synology NAS-DR-DC	Network Attached Storage	Remote Data Center	10.5.1.223

IoT

Device	Device Type	Location	IP Address
Devio BiAmp	Internet of Things (IoT)	Unknown	10.5.1.146
Devio BiAmp	Internet of Things (IoT)	Unknown	10.5.1.177
TME Thermometer	Internet of Things (IoT)	Unknown	10.5.1.242

Virtual Machines

Device	Device Type	Location	IP Address
ACRONIS	Acronis Appliance		10.5.2.50
DATASVR	File Server		10.5.1.215
BK2	Veeam Backups		10.5.1.228
BK3	DC, Symprex, AAD Connect, Vray		10.5.1.6
DC	DC FSMO Roles		10.5.1.217
EXCH2013	Exchange Server		10.5.1.218
FS01	Archive File Server		10.5.1.208
LIC	License Server, LMTools, Zoo		10.5.1.206
PRINT	Print Server		10.5.1.213
SHAR2 (Not Used)	Legacy Sharepoint Server		10.5.1.222
WDS (Not Used)	Windows Deployment Services		10.5.1.212
WSUS	DC, WSUS, Lansweeper & PDQ		10.5.1.35
TIMESHEET	Timesheet System & Shares		10.5.1.220
VC01	vCenter Appliance		10.5.2.99
VCENTER (Not Used)	Legacy vCenter		10.5.1.226

Mitel

Device	Device Type	Location	IP Address
Mitel vPBX	Virtual Machine	Mitel Cloud	198.51.100.83
Mitel vPBX	Virtual Machine	Mitel Cloud	198.51.100.84
Mitel vPBX	Virtual Machine	Mitel Cloud	198.51.100.85
Mitel vPBX	Virtual Machine	Mitel Cloud	198.51.100.86

PHYSICAL SERVERS

There are 3 Dell enterprise server running in the Contoso Ltd production network. 2 of them hosts the VMware Hypervisor cluster and the 3rd is running as a multipurpose Windows Server. Since they are all critical for production their specifications are documented in the following pages.

Dell PowerEdge R320 | BK3

[Name]

Netbios Name	bk3
FQDN Name	bk3.contoso.com.au
Model	PowerEdge R320

[Motherboard]

CPU Type	Xeon(R) CPU E5-2420 v2 @ 2.20GHz, 6 Core(s), 12 Logical Processor(s)
Processor Sockets	2
Cores per Socket	6
Logical Processors	12
Hyperthreading	Active
Number of NICs	4

[Operating System]

OS Name	Windows Server 2012 R2
OS Version	Version 6.3 Build 9600
OS Service Pack	R2

[Memory]

Physical Memory:	32 GB
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[Network Configuration]

Network Adapter	2 x Broadcom NetXtreme 1 Gigabit Ethernet Adapters
Interface Type	Gigabit Ethernet
Connection Speed	1000 Mbps
IP / Subnet Mask	10.5.1.6 / 255.255.255.0
Default Gateway	10.5.1.252
DHCP	N/A
DNS1	10.5.1.6
DNS2	8.8.8.8

Dell PowerEdge R630 | MEL-ESX-01

[Name]

Netbios Name	mel-esx-01
FQDN Name	mel-esx-01.contoso.com.au
Model	PowerEdge R630

[Motherboard]

CPU Type	Xeon(R) CPU E5-2650 v3 @ 2.30GHz, 6 Core(s), 12 Logical Processor(s)
Processor Sockets	2
Cores per Socket	10
Logical Processors	40 (count is x2 due to enabled Hyperthreading)
Hyperthreading	Active
Number of NICs	4

[Operating System]

OS Name	VMware ESXi
OS Version	6.0.0.5572656
OS Service Pack	N/A

[Memory]

Physical Memory:	128 GB
------------------	--------

[Network Configuration]

Network Adapter	2 x Broadcom Qlogic 57800 10 Gigabit Ethernet Adapters
Interface Type	Gigabit Ethernet
Connection Speed	10000 Mbps

Network Adapter	2 x Broadcom Qlogic 57800 1 Gigabit Ethernet Adapters
Interface Type	Gigabit Ethernet
Connection Speed	1000 Mbps

IP / Subnet Mask	10.5.2.100 / 255.255.255.0
Default Gateway	10.5.2.254
DHCP	N/A
DNS1	10.5.1.217
DNS2	10.89.43.22

Dell PowerEdge R630 | MEL-ESX-02

[Name]

Netbios Name	mel-esx-02
FQDN Name	mel-esx-02.contoso.com.au
Model	PowerEdge R630

[Motherboard]

CPU Type	Xeon(R) CPU E5-2650 v3 @ 2.30GHz, 6 Core(s), 12 Logical Processor(s)
Processor Sockets	2
Cores per Socket	10
Logical Processors	40 (count is x2 due to enabled Hyperthreading)
Hyperthreading	Active
Number of NICs	4

[Operating System]

OS Name	VMware ESXi
OS Version	6.0.0.5572656
OS Service Pack	N/A

[Memory]

Physical Memory:	128 GB
------------------	--------

[Network Configuration]

Network Adapter	2 x Broadcom Qlogic 57800 10 Gigabit Ethernet Adapters
Interface Type	Gigabit Ethernet
Connection Speed	10000 Mbps

Network Adapter	2 x Broadcom Qlogic 57800 1 Gigabit Ethernet Adapters
Interface Type	Gigabit Ethernet
Connection Speed	1000 Mbps

IP / Subnet Mask	10.5.2.101 / 255.255.255.0
Default Gateway	10.5.2.254
DHCP	N/A
DNS1	10.5.1.217
DNS2	10.89.43.22

NETWORKING

The following sections contains important information about the Contoso Ltd core network configuration. Everything from the IP Address Space, Routing, DHCP, DNS, VLANs and a High-Level Topology are covered to provide an overview of the current network design and to help us understand what potential changes could improve the health of the network moving forward.

IP Address Space

Below table contains a summary of the IP address ranges already configured. Not all the ranges are in use or have any purpose. This is most likely the result of work initiated, but not completed in an attempt to improve the current address space allocation.

Server Network [Default]	Subnet Mask	CIDR Notation	VLAN
10.5.1.0	255.255.255.0	/24	1
Client Network [Default]	Subnet Mask	CIDR Notation	VLAN
10.5.1.0	255.255.255.0	/24	1
Wi-Fi Network [Default]	Subnet Mask	CIDR Notation	VLAN
10.5.1.0	255.255.255.0	/24	1
Management Network	Subnet Mask	CIDR Notation	VLAN
10.5.2.0	255.255.255.0	/24	2
vMotion Network	Subnet Mask	CIDR Notation	VLAN
10.0.70.0	255.255.255.0	/24	70
Voice Network	Subnet Mask	CIDR Notation	VLAN
10.5.10.0	255.255.255.0	/24	80
DMZ Network	Subnet Mask	CIDR Notation	VLAN
N/A	N/A	N/A	99
Noname Network	Subnet Mask	CIDR Notation	VLAN
10.1.1.0	255.255.255.0	/24	100
Public Network	Subnet Mask	CIDR Notation	VLAN
172.16.1.0	255.255.255.0	/24	101

Routing and Gateways

No dynamic routing protocols are currently deployed in the network, instead a range of static routes are defined on most of the layer 3 devices.

As seen in the table the default route destination or gateway is represented by a certain color associated with the relevant layer 3 device allowing us to quickly identify where the next-hop destination is and what type of device it belongs to.

In this context it should be mentioned that all the Access Layers switches are configured as a Layer 2 devices or only supports Layer 2 functionality, as a result none of them are included in the table, except for the Cisco SG300 Top switch. Refer to the [Layer 2 and 3 Devices](#) section to get a complete overview of the Layer 2 switches or to the [High-Level Topology](#) to see where they are in use.

Not all routes are relevant or up-to-date and certain hops are no longer routable, however all configured routes are documented for completeness' sake and to help us make decisions about future design changes.

Device	Destination	Gateway	Interface	Type	Metric	Distance
1424 OneAccess SHDSL Unknown IP	Managed by 3rd party	N/A	N/A	N/A	N/A	N/A
Cisco 881 10.5.1.211	Managed by 3rd party	N/A	N/A	N/A	N/A	N/A
HP ProCurve 5406zl 10.5.1.240	0.0.0.0 / 0.0.0.0	10.5.1.252	VLAN1	Static	1	1
	10.1.1.0 / 255.255.255.0	VLAN100	VLAN100	Connected	1	0
	172.16.1.0 / 255.255.255.0	VLAN101	VLAN101	Connected	1	0
	10.5.1.0 / 255.255.255.0	DEFAULT_VLAN	VLAN1	Connected	1	0
HP ProCurve 5412Rzl2 10.5.1.210	0.0.0.0 / 0.0.0.0	10.5.1.252	VLAN1	Static	1	1
	10.1.1.0 / 255.255.255.0	10.5.1.240	VLAN1	Static	1	1
	10.89.43.0 / 255.255.255.0	10.5.1.211	VLAN1	Static	1	1
	172.16.1.0 / 255.255.255.0	10.5.1.240	VLAN1	Static	1	1
	10.5.1.0 / 255.255.255.0	DEFAULT_VLAN	VLAN1	Connected	1	0
	10.5.2.0 / 255.255.255.0	Management	VLAN2	Connected	1	0
	10.5.100.0 / 255.255.255.0	10.5.1.211	VLAN1	Static	1	1
Sophos XG 210 10.5.1.252	Internet (OR out of network)	198.51.100.10	Port 2	N/A	0	N/A
	10.5.0.0 / 255.255.0.0	10.5.1.210	Port 1	N/A	0	N/A
	10.0.0.0 / 255.0.0.0	10.5.1.210	Port 1	N/A	0	N/A
	10.5.100.0 / 255.255.255.0	10.5.1.237	Port 1	N/A	0	N/A
Cisco Switch SG300 10.5.1.237	N/A	N/A	N/A	N/A	N/A	N/A

Firewall

(Work in progress / example only).

Interfaces

DNS

DHCP

Network Zones

Routing Gateways

Authentications Servers

Rules and Policies

Host and Services

DHCP

There are multiple DHCP instances running in the network, the first service is running on the DC and hosts the 10.5.1.0/24 scope accommodating most clients and wireless devices.

A range of additional scopes are also configured on the DC DHCP service but not all are relevant or in use, however all information is documented for completeness' sake and to help us make decisions about future design changes.

DHCP Service: LDC.contoso.com.au [10.5.1.217]					
Scope Name [10.5.1.0]	Type	Start Address	End Address	Subnet Mask	Other
Scope Contoso Address Pools	DHCP IPv4 Scope	10.5.1.2	10.5.1.202	/24	8 Hours
	IP Pool	10.5.1.2	10.5.1.202	/24	Distribution
	IP Pool	10.5.1.6	10.5.1.6	/24	Excluded
	IP Pool	10.5.1.35	10.5.1.35	/24	Excluded
	IP Pool	10.5.1.41	10.5.1.41	/24	Excluded
	IP Pool	10.5.1.100	10.5.1.101	/24	Excluded
	IP Pool	10.5.1.140	10.5.1.140	/24	Excluded
Scope Name [10.5.2.0]	Type	Start Address	End Address	Subnet Mask	Other
Scope Management	DHCP IPv4 Scope	10.5.2.50	10.5.2.80	/24	1 Day
Address Pools	IP Pool	10.5.2.50	10.5.2.80	/24	Distribution
Scope Name [172.16.1.0]	Type	Start Address	End Address	Subnet Mask	Other
Scope Public Wireless	DHCP IPv4 Scope	172.16.1.1	172.16.1.199	/24	8 Hours
Address Pools	IP Pool	172.16.1.1	172.16.1.199	/24	Distribution
Scope Name [10.1.1.0]	Type	Start Address	End Address	Subnet Mask	Other
Scope Wireless	DHCP IPv4 Scope	10.1.1.1	10.1.1.199	/24	8 Hours
Address Pools	IP Pool	10.1.1.1	10.1.1.99	/24	Distribution
Scope Options [10.5.1.0]	Vendor	Value			Class
003 Router	Standard	10.5.1.252			None
006 DNS Servers	Standard	10.5.1.217 + 6			None
015 DNS Domain Name	Standard	contoso.com.au			None
Scope Options [10.5.2.0]	Vendor	Value			Class
003 Router	Standard	10.5.2.254			None
006 DNS Servers	Standard	10.5.1.217			None
015 DNS Domain Name	Standard	contoso.com.au			None
Scope Options [172.16.1.0]	Vendor	Value			Class
003 Router	Standard	10.5.2.252			None
006 DNS Servers	Standard	8.8.8.8			None
015 DNS Domain Name	Standard	contoso.com.au			None
Scope Options [10.1.1.0]	Vendor	Value			Class
003 Router	Standard	10.5.2.252			None
006 DNS Servers	Standard	10.5.1.217 + 6			None
015 DNS Domain Name	Standard	contoso.com.au			None

The second service is running on the Sophos XG 210 Firewall and host the 10.5.10.0/24 scope accommodating all the Mitel handsets.

Again, a range of additional scopes are also configured on the Sophos XG 210 DHCP service but not all are relevant or in use, however all information is captured for completeness' sake and to help us make decisions about future design changes.

DHCP Service: Sophos XG 210 [10.5.10.252]					
Scope Names [10.5.10.0]	Type	Start Address	End Address	Subnet Mask	Other
Mitel_Handsets	DHCP IPv4 Scope	10.5.10.50	10.5.10.200	/24	24 Hours
Address Pools	IP Pool	10.5.10.50	10.5.10.200	/24	Distribution
Scope Names [10.255.0.1]	Type	Start Address	End Address	Subnet Mask	Other
GuestAP	DHCP IPv4 Scope	10.255.0.2	10.255.0.254	/24	24 Hours
Address Pools	IP Pool	10.255.0.2	10.255.0.254	/24	Distribution
Scope Options [10.5.1.0]	Vendor	Value			Class
003 Router	Standard	10.5.10.252			None
006 DNS Servers	Standard	1.1.1.1 + 8.8.8.8			None
Scope Options [10.255.0.1]	Vendor	Value			Class
003 Router	Standard	10.255.0.1			None
006 DNS Servers	Standard	10.255.0.1			None

DNS

As seen in below table there are multiple Windows DNS Servers registered as per current Active Directory Domain Service records. 2 of them specifically, DC and BK3 are active and are hosting the relevant zone files. 3 of them are impacted by degraded services or are not contactable.

Not all records are relevant or in use, however all information is captured for completeness' sake and to help us make decisions about future design changes.

DNS Service: contoso.com.au					
DNS name	Type	IP Address	Domain	Forwarders	Status
dc.contoso.com.au	Windows NS	10.5.1.217	contoso.com.au	198.51.100.35 198.51.100.36	Online
DNS name	Type	Address	Domain	Forwarders	Status
bk3.contoso.com.au	Windows NS	10.5.1.6	contoso.com.au	198.51.100.22 198.51.100.2 8.8.8.8 8.8.4.4	Online
DNS name	Type	Address	Domain	Forwarders	Status
wsus.contoso.com.au	Windows NS	10.5.1.35	contoso.com.au	N/A	No Service
DNS name	Type	Address	Domain	Forwarders	Status
dc-dr.contoso.com.au	Windows NS	10.5.1.216	contoso.com.au	N/A	Offline
DNS name	Type	Address	Domain	Forwarders	Status
dcdr.contoso.com.au	Windows NS	Unknown	contoso.com.au	N/A	Offline
Zone files	SOA Record				
_msdcs.contoso.com.au	IN SOA dc.contoso.com.au. hostmaster.contoso.com.au.				
www.contoso.com.au	IN SOA dc.contoso.com.au. hostmaster.contoso.com.au.				
outbound_DNS	IN SOA dc.contoso.com.au. hostmaster.contoso.com.au.				
contoso.com.au	IN SOA dc.contoso.com.au. hostmaster.contoso.com.au.				
www.contonso.com.au	IN SOA dc.contoso.com.au. hostmaster.contoso.com.au.				

VLANs

In the following section a range of different VLAN maps are presented to help us get an overview of which network devices are participating in each VLAN and whether they have an active role in the virtual LAN or are simply configured as VLAN ready.

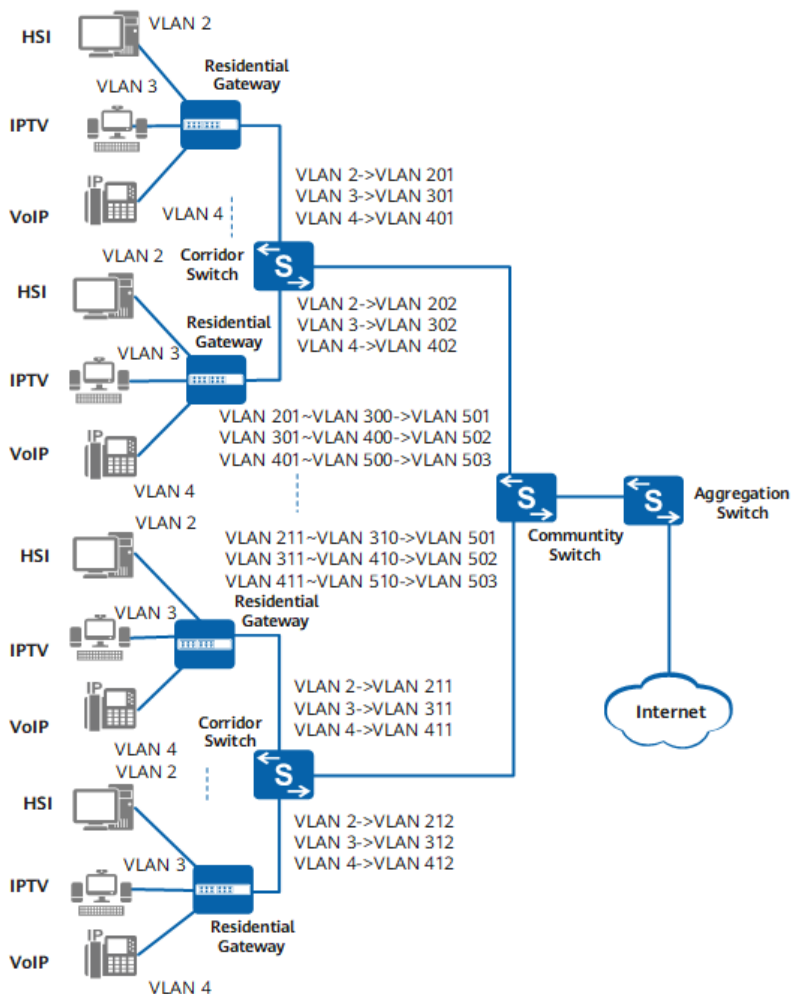
Looking at the VLAN configuration for the Contoso Ltd network and for a moment ignoring the Hypervisor and Management networks there are essentially only 2 priority VLAN's in use, namely VLAN 1 (default) covering both Servers, Clients and Wireless Devices and VLAN 80 covering the Mitel Handsets. The VLAN 1 (default) network is associated with a Private Class C range allowing only 254 hosts, which is an interesting choice considering the broad range of devices it covers.

Looking at the Management VLAN or VLAN 2 we see a select but small range of both Layer 2 and 3 devices, the ESX hosts and a couple of VMs associated with this segment. Again, an interesting choice since one would think all Layer 2 and 3 devices would be associated with this segment.

Finally looking at the Hypervisor VLANs which includes: VLAN 70, 99 & 101 we see a range of VLANs configured but not in use with the exception of VLAN 70 which is used for vMotion. This is most likely the result of work initiated, but not completed or possibly legacy network configuration that is no longer active.

VLAN Diagram

(Work in progress / example only).



VLAN Mapping

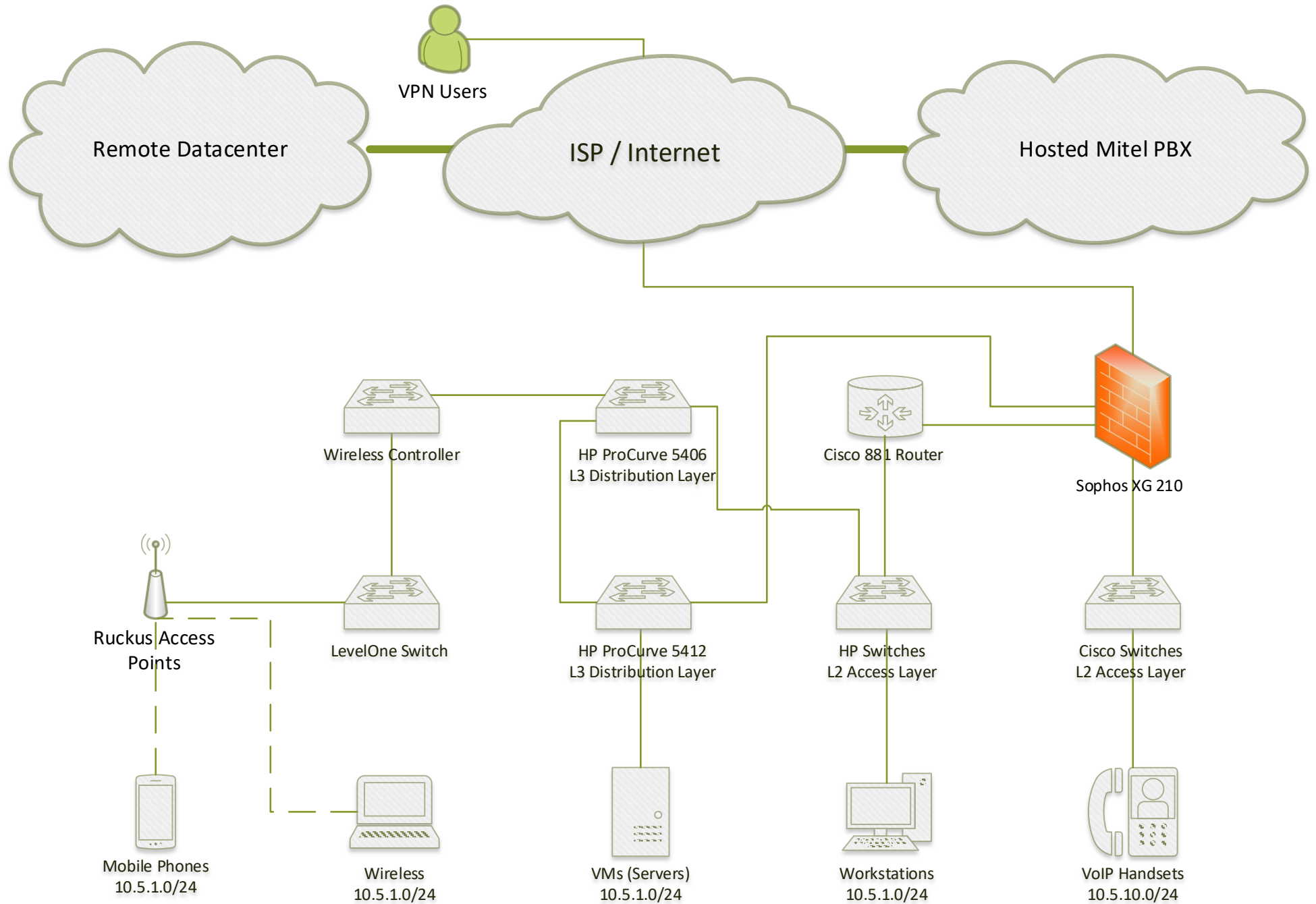
The following VLAN table provides a per-device-VLAN cross reference to be able to determine the impact of adding or removing VLANs to / from the core network and how it could potentially prevent certain devices from communicating after such action.

To keep the output reasonably simple the table does not show tagged or untagged VLAN data and it only contains Layer 2 and 3 devices plus Production servers to present the VLAN information in a minimalistic manner.

- Green highlights represent a VLAN being configured on the specific device (but does not necessarily indicate participation in the VLAN network).
- Red highlights represent a VLAN not being configured on the specific device.
- Yellow highlights represent a device not currently accessible.

VLAN Name	Default	Management	vMotion	Voice	DMZ	N/A	Public
VLAN Enabled / IP	VLAN 1	VLAN 2	VLAN 70	VLAN 80	VLAN 99	VLAN 100	VLAN 101
Dell PE R630 MEL-ESX-01	10.5.1.230	10.5.2.100	10.0.70.10				
Dell PE R630 MEL-ESX-02	10.5.1.231	10.5.2.101	10.0.70.11				
HP ProCurve 5412	10.5.1.210	10.5.2.254					
HP ProCurve 5406	10.5.1.240					10.1.1.254	172.16.1.254
Cisco SG300 Top	10.5.1.237	10.5.2.30		10.5.10.10			
Cisco SG300 Middle	10.5.1.219	10.5.2.31		10.5.10.11			
Cisco SG350 Bottom	10.5.1.221	10.5.2.32		10.5.10.12			
Linksys SRW2048	10.5.1.221						
HP OC 1820 Top	10.5.1.241						
HP OC 1820 Bottom	10.5.1.236						
HP ProCurve 1800 DS1	10.5.1.233						
HP ProCurve 1800 DS2	10.5.1.234						
HP ProCurve 2824 DS3	10.5.1.232						
Level 1 GEP-0950	10.5.1.1						
VNXe 3200 SP-A		10.5.2.90					
VNXe 3200 SP-B		N/A					
Dell PE R230 LBK3	10.5.1.6						
Dell xStack DES-3200-10	Update						

High-Level Topology



NETWORK DEVICE CONNECTIVITY

A couple of diagram types are presented in the following section

- The Device Interconnections diagram
- The Device Traffic Flow diagram(s)

The Interconnection diagram has the purpose of visually presenting all cabling between devices, initially this may seem a bit excessive to present the exact wiring between each connected device, however this type of diagram is very useful when network design changes are to be considered.

Take a look at [Device Interconnections](#) and let's clarify by using a couple of examples.

1. If we decided to disconnect the top Cisco SG300 access layer switch we would know that the entire Voice network would go down, because this switch acts not only as a Voice access layer switch but also acts as primary uplink switch for all the other Voice access layer switches.
2. If on the other hand if we decided to disconnect the Bottom Cisco SG350 access layer switch we would know that only about 1/3 of the Voice network would go down, as this switch acts only as a Voice access layer switch for about 1/3 of the Mitel Handset and does not serve as an Uplink for other switches.
3. We can also see there is a direct connection between the HP 5406 ProCurve and the Huawei S5320 which is rather unusual as this is bypassing the Firewall and is possibly a direct link to the outside world. This could be the link to the Remote Datacentre, but this could also be an outdated patch point.

The Traffic Flow diagrams have the purpose of visually representing traffic travelling in and out of the network based on device type. Again, this is useful not only to see the traffic type flowing in and out of the network but also to observe the actual wiring required for that particular type of traffic to work.

Again, let's clarify by using a couple of examples.

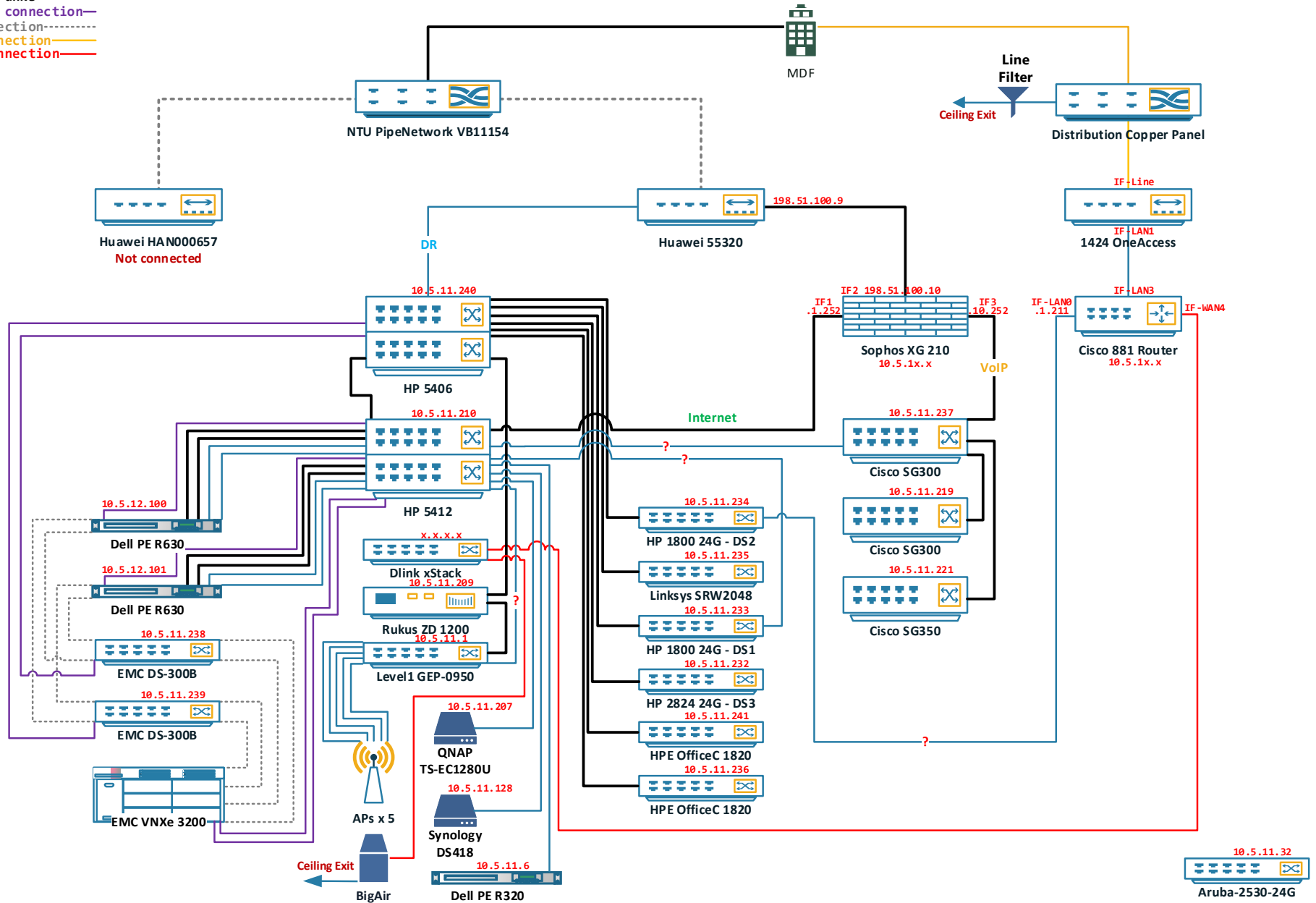
1. Looking at the [Workstations Traffic Flow](#) we know that traffic flow from Workstations to the Internet is taking the following route: Workstation <-> Specific Access Layer switch <-> HP 5406 <-> HP 5412 <-> Firewall IF1 <-> Huawei S5320 <-> NTU <-> ISP.
2. Looking at [VoIP Traffic Flow](#) we know that the traffic flow from the Voice network is completely bypassing the Distribution switches and are going directly out of the network through IF2 on the Sophos Firewall.

Above examples shows why the interconnectivity / traffic flow diagrams are useful as this information could not be deduced from the other network documentation.

Device Interconnections

Legend

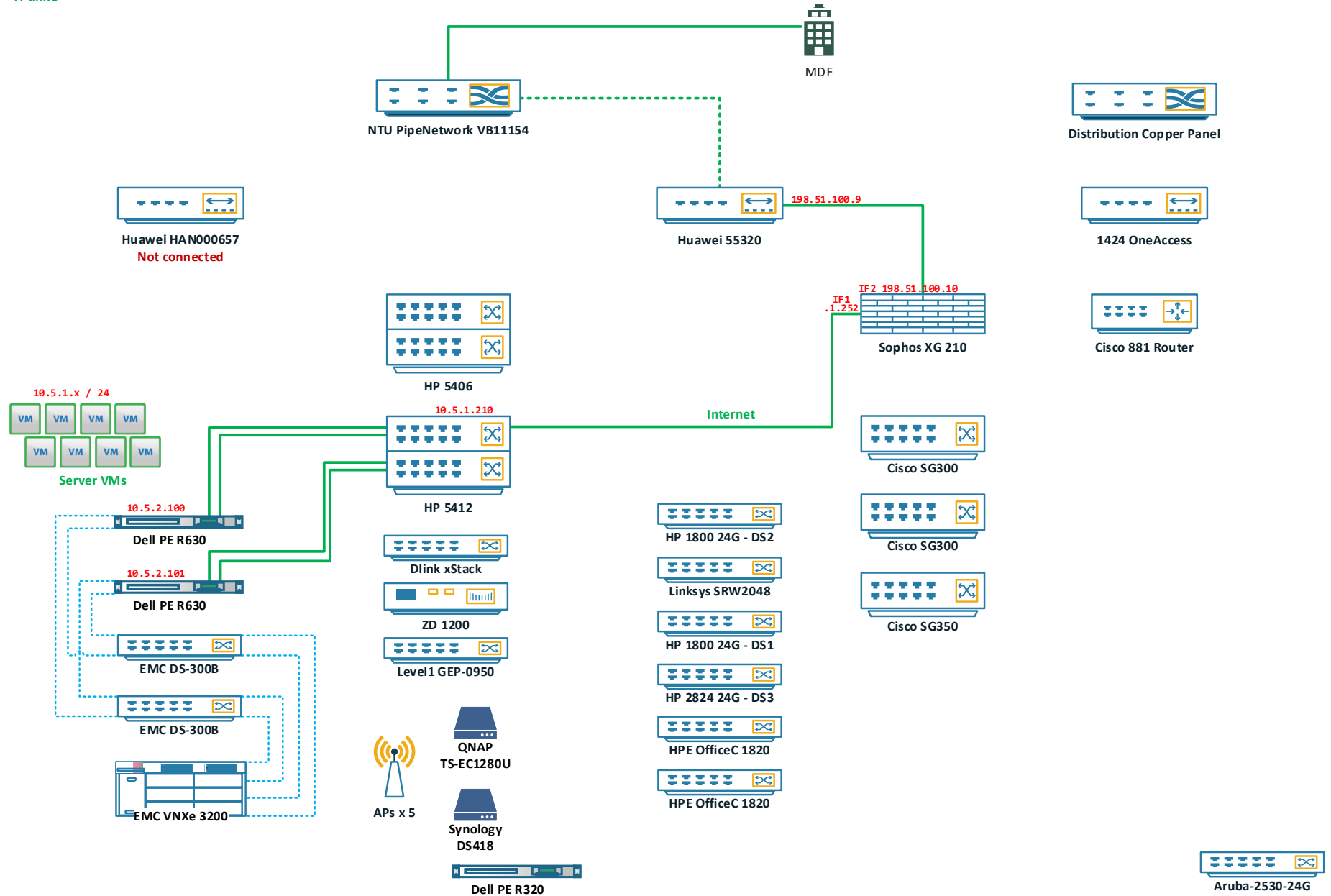
- Data connection—————
- Uplink / Trunks—————
- Management connection—
- Fibre connection-----
- Copper connection————
- Unknown connection————



Servers Traffic Flow

Legend

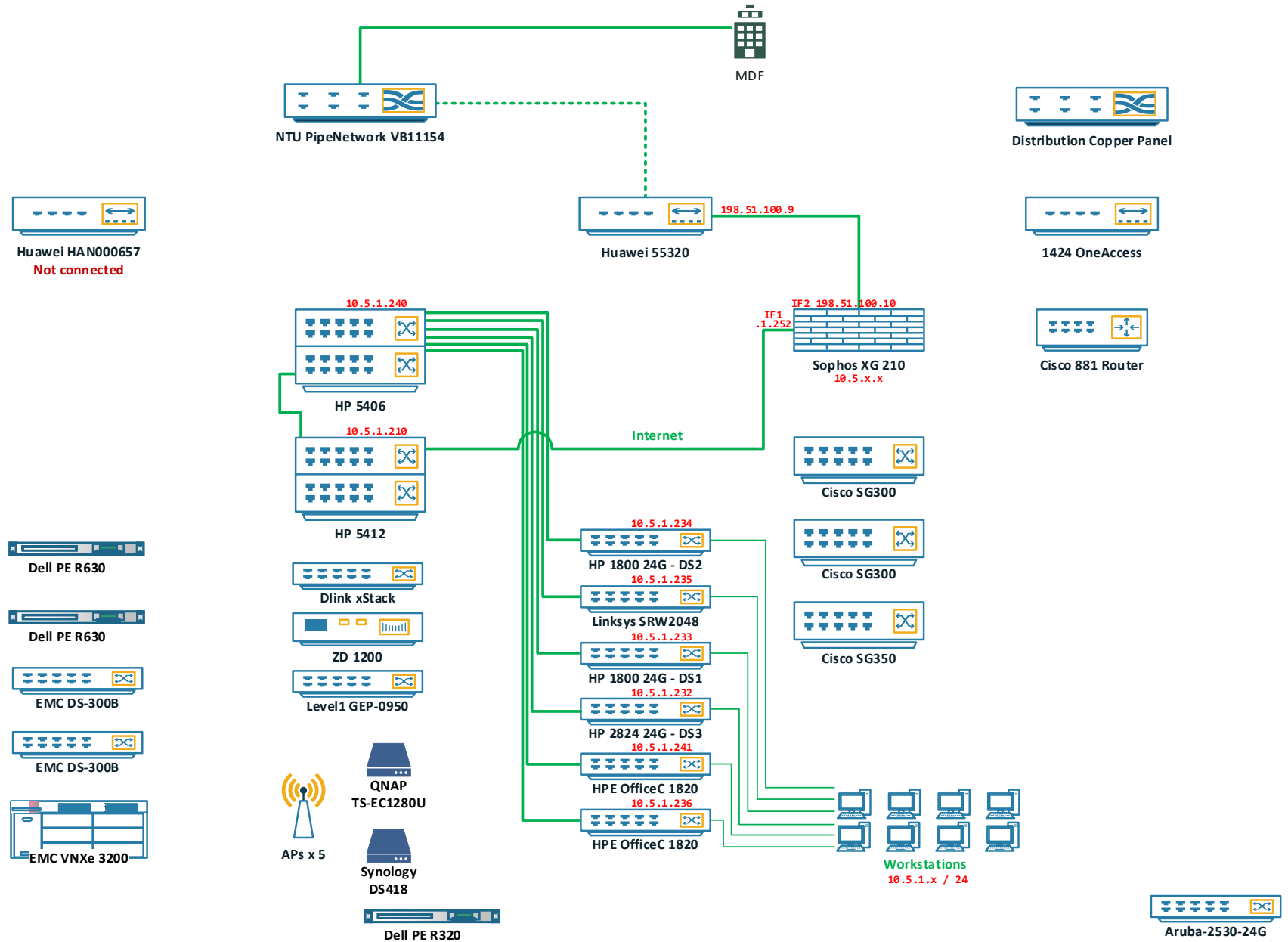
- Data connection-
- Uplink / Trunks-



Workstations Traffic Flow

Legend

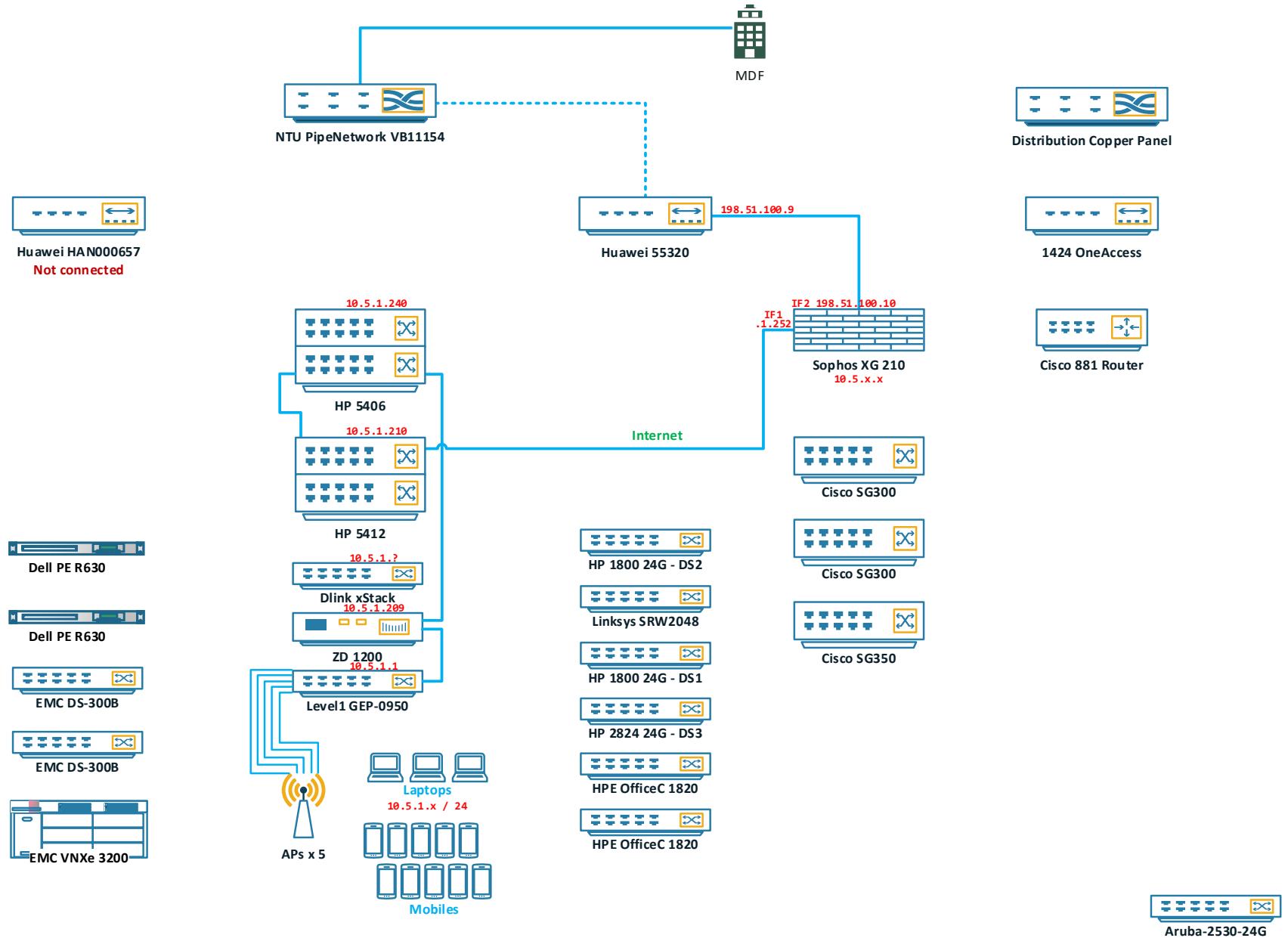
- Data connection
- Uplink / Trunks



Wireless Traffic Flow

Legend

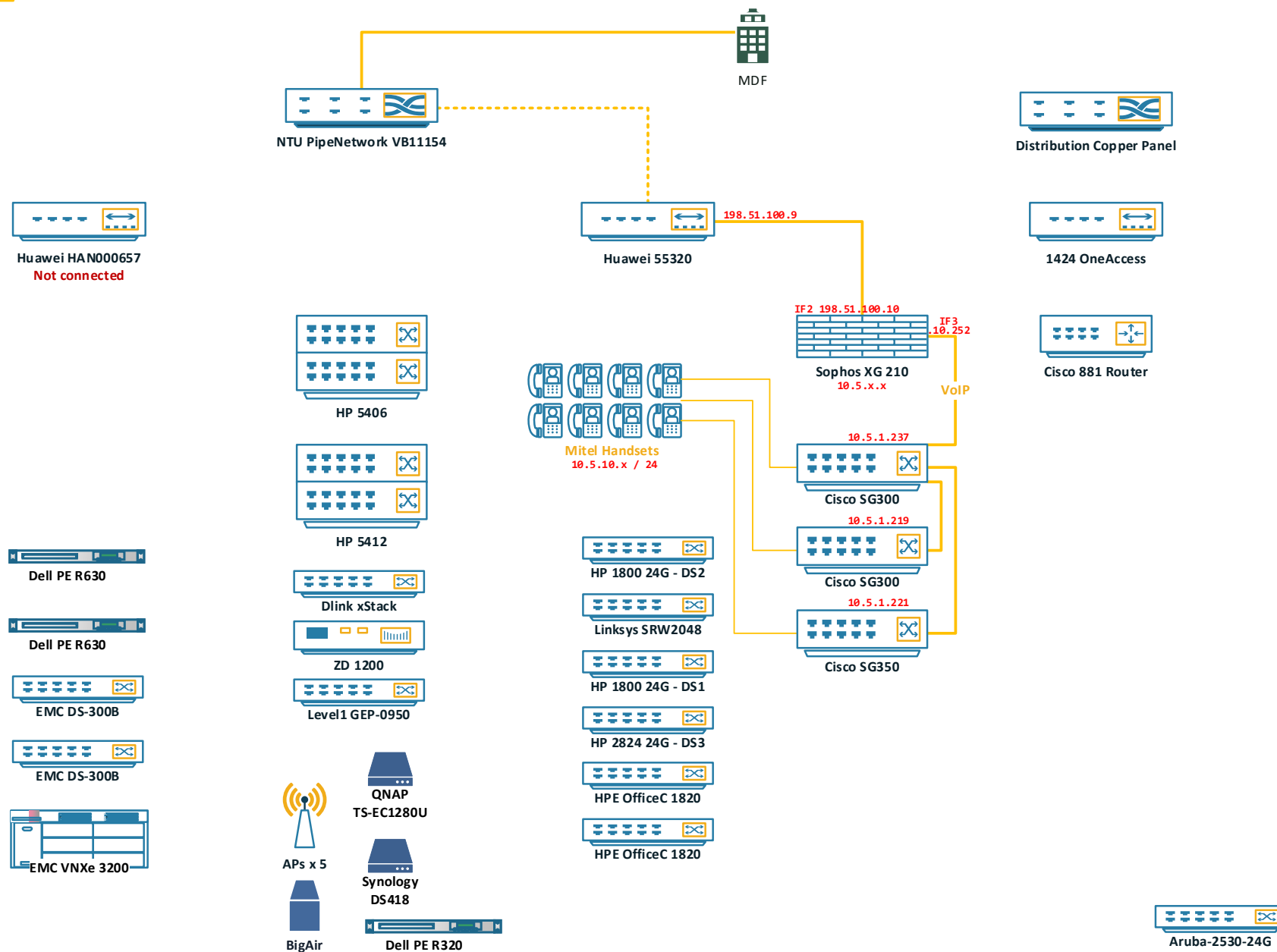
- Data connection
- Uplink / Trunks



VoIP Traffic Flow

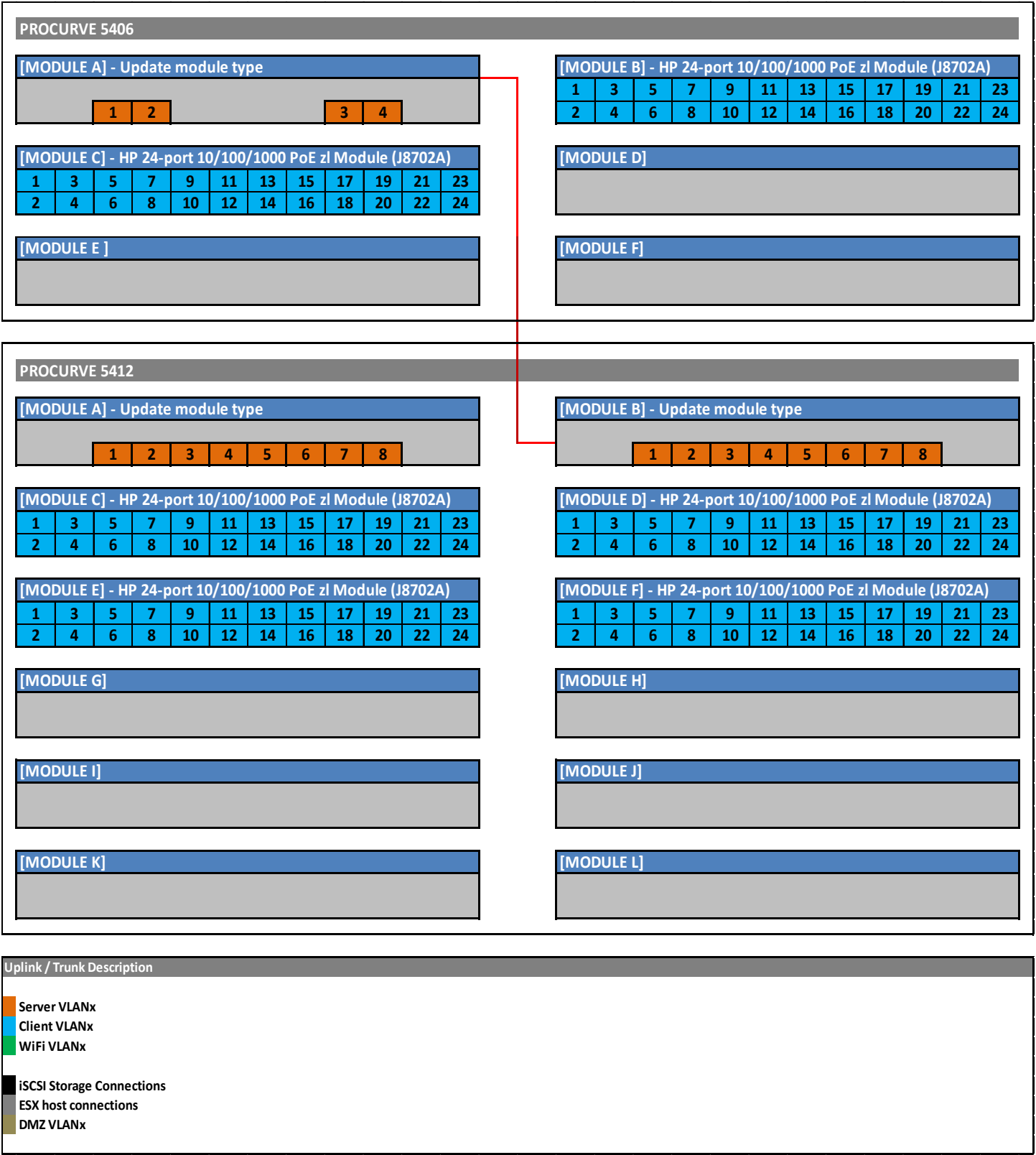
Legend

- Data connection
- Uplink / Trunks



UPLINKS AND TRUNKS

(Work in progress / example only).



SAN STORAGE OVERVIEW

(Work in progress / example only).

VNXe 3200 Storage Configuration

[Disk Configuration](#)

[Array Specifications](#)

[Network Settings](#)

[Volumes](#)

[Host Initiator Groups](#)

Dell PowerEdge R630 Host Configuration

[Network Configuration](#)

[vSphere Configuration](#)

[vSphere Network Configuration](#)

VMware vCenter Server Configuration

[vCenter Server Specifications](#)

[vCenter Server IP Information](#)

[Sites and Datacentres](#)

EMC Brocade Fiber Switch Configuration

[VLAN Configuration](#)

[Port Configuration](#)

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